

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. in CSE Final Examination, 2022
CSE-1101: Fundamentals of Computers and Computing

Total Marks: 70

Time: 3 Hours

Answer any 5 (Five) of the following Questions

1.
 - a) Describe details about how information is processed in a computer system. 4
 - b) A program is running on a specific machine (CPU) with the following parameters: 4
 - i) Total instructions executed = 10^7
 - ii) Average CPI = 2.5 cycles per instruction.
 - iii) CPU clock rate = 200MHz (clock cycle = $1/\text{clock rate}$).
 Find the execution time for this program.
 - c) Compare between application and system software with proper example. 4
 - d) What are data and data processing? 2
2.
 - a) What do you mean by ROM, PROM, EPROM and EEPROM? 4
 - b) If the disk is rotating at 5400 RPM, what are the data transfer rate and average access time? The specifications are as following: 6
 - i) Number of surfaces = 16
 - ii) Number of tracks per surface = 128
 - iii) Number of sectors per track = 256
 - iv) Number of bytes per sector = 512 bytes
 - v) Average seek time = 25 msec.
 - c) Why SRAM and DRAM are called static and dynamic respectively? 4
3.
 - a) What are UNICODE and ASCII? 2
 - b) Convert the numbers as following: 4
 - i) $11001101_2 = ?_8$
 - ii) $F9DE_{16} = ?_{10}$
 - iii) $7652_8 = ?_{16}$
 - iv) $789.1256_{10} = ?_2$
 - c) Let $A = 63$ and $B = 27$. Perform $A \& B$, A/B and $A \wedge B$. 6
 - d) Make comparison between server and workstation. 2
4.
 - a) Write short notes on the following Operating Systems: 6
 - I. Single-User/Single-Tasking OS.
 - II. Single-User/Multitasking OS.
 - III. Multi-User/Multitasking OS.
 - b) Explain the basic functional operations of a Microprocessor. 4
 - c) Write a comparative study among Database, Transnational and Network File Systems. 4
5.
 - a) Mention some components needed to establish a computer networking. 2
 - b) Distinguish between LAN, MAN and WAN. 4
 - c) Assume that, two devices connected to a network. Will they carry identical IP addresses? Give an explanation in favor of your opinion. 2
 - d) Draw the diagram and write down the characteristics of Star, Mesh and Tree Topology. 6
6.
 - a) What is a Program? Write down the basic structure of a C program. 3
 - b) Write down some rules for naming variables in C. 3
 - c) Write an Algorithm and draw a Flowchart to find the largest number among three integer numbers. 4
 - d) Write a C program to find out the Area of a Triangle using the length of its arms. 4
7.
 - a) What is SSD? How a SSD relates to a SDRAM or HDD? 4
 - b) Describe the factors that effects to a processor's speed. 6
 - c) Write a short note on Flash memory and Cache memory. 4